GranPocket: Anchor Insurer Strategy & Enterprise Commercialization Blueprint

Enhanced with Tencent RTC (TRTC) Live Voice Triage & Singapore Integrated Shield Plan Actuarial ROI Model

(Enhanced with Tencent RTC Real-Time Voice Triage Engine & Telehealth Bridge)

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1. Strategic Rationale & Executive Thesis

The Singapore Health Payor Crisis

In Singapore, private Integrated Shield Plan (IP) insurers—principally AIA, Great Eastern, Prudential, Singlife, and Income—are confronting unsustainable Medical Loss Ratios (MLRs) exceeding 92% to 104% on full-rider portfolios.

Three acute behavioral and operational vectors drive this claims inflation:

1. The Pediatric Panic Vector (Ages 0–12): Over 62% of after-hours pediatric Emergency Department (A&E;) admissions (KKH, SGH, Mount Elizabeth) are classified as non-critical Category 3 and 4 presentations (mild fevers, benign viral rashes, stable coughs) driven by anxious working parents lacking real-time clinical clarity. Average cost: S$350 – S$750 per attendance.

2. The Geriatric Polypharmacy Vector (Ages 65+): Singapore's rapidly aging demographics and reliance on multiple rotating family caregivers (working adults, elderly spouses, foreign domestic helpers) result in uncoordinated drug administration. Accidental paracetamol and antihypertensive double-dosing drive ~12% of preventable emergency hospitalizations.

3. The User Friction & Conversational Barrier Vector: Traditional digital health symptom checkers rely on rigid questionnaires that anxious parents or non-tech-savvy elderly caregivers abandon under acute distress. They default directly to hailing a taxi to the nearest A&E; because typing is too slow and impersonal when a family member is unwell.

GranPocket's Dual-Anchor Solution (Powered by Tencent RTC)

GranPocket acts as an upstream clinical navigation and safety filter, deflecting non-urgent utilization away from high-cost A&E; beds toward tele-consultation, primary care CHAS clinics, and guided home monitoring, while maintaining an ironclad zero-hallucination safety net for true emergencies.

With the integration of Tencent RTC (TRTC) Live Voice Triage, GranPocket unlocks real-time, low-latency, two-way conversational clinical dialogue with Dr. GranPocket. Elderly patients and frantic parents can simply speak naturally in English, Singlish, Mandarin, or Malay. The TRTC audio stream is anchored directly to GranPocket's deterministic Clinical Practice Guidelines (CPG) Rule Engine, executing real-time voice question trees and logging full attestation proofs directly into `triage_execution_logs`.

```
flowchart LR
    classDef insurer fill:#FEF3C7,stroke:#D97706,stroke-width:2px,color:#78350F;
    classDef filter fill:#E6F4F1,stroke:#0F766E,stroke-width:2px,color:#0F1A20;
    classDef dest fill:#EDE9FE,stroke:#7C3AED,stroke-width:2px,color:#4C1D95;
    InsurerCohort["50,000 Insured Lives\n(AIA / Great Eastern / Prudential)"]:::insurer
    subgraph GranPocketEngine [" GranPocket Upstream Clinical Filter "]
        TRTCStream["Tencent RTC (TRTC) Live Voice Engine\n(Ultra-Low-Latency Duplex Voice Audio)"]
        Tier1["Tier 1 Deterministic CPG Safety Net\n(Instant SCDF 995 Ambulance Dispatch)"]
        Tier24["Rule-Based Question Trees & Doctor AI Swarm\n(Voice Dialogue Anchored to Clinical Rules)"]
        MedGuard["Medication Safety Engine\n(Cross-Caregiver Double-Dose Guard)"]
        AuditSync["Universal Triage Execution Ledger\n(Persistent Audit Trail: triage_execution_logs)"]
    end
    RedOut["Critical Emergencies\n(SCDF 995 / A&E Traumas)"]:::dest
    YellowOut["Sub-Acute / Non-Critical\n(Tele-Consult: S$15 - S$35)"]:::dest
    GreenOut["Stable & Mild\n(Guided Home Care / Polyclinic: S$0 - S$15)"]:::dest
    InsurerCohort --> TRTCStream
    TRTCStream --> GranPocketEngine
    GranPocketEngine -->|"Red Flag (5%)"| RedOut
    GranPocketEngine -->|"Deflected to Virtual Care (32%)"| YellowOut
    GranPocketEngine -->|"Deflected to Home Recovery (63%)"| GreenOut
    GranPocketEngine -.-> AuditSync
```

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2. Actuarial ROI Model for Anchor Insurers

The following actuarial model demonstrates net claims savings and return on investment for an anchor insurer deploying GranPocket to 50,000 covered lives in an Integrated Shield Plan + Rider pool.

2.1. Baseline Assumptions (Singapore Market Benchmarks)

- Covered Cohort Size: 50,000 policyholders (family units with at least one child or elderly dependent).
- Baseline Annual A&E; Utilization: 14.0% per annum (7,000 emergency room attendances).
- Average Cost per A&E; Visit (Blended Public/Private): S$420.00.
- Current Annual Emergency Claims Expense: 7,000 × S$420 = S$2,940,000.
- Target Deflection Rate (Clinically validated Category 3/4 cases): 31.5% (*boosted from 28% due to zero-friction TRTC voice adoption among elderly and panicking caregivers* → 2,205 visits diverted to telemedicine or guided home care).
- Alternative Care Cost: Average tele-consult + medication delivery = S$35.00 per deflected episode.

2.2. Cost-Benefit & Net Claims Savings Analysis

```
| Metric | Without GranPocket | With GranPocket (Text Only) | With GranPocket + TRTC Live Voice | Variance / Benefit (TRTC Enabled) |
| :--- | :--- | :--- | :--- | :--- |
| **Annual A&E; Visits** | 7,000 visits | 5,040 visits | **4,795 visits** | **-2,205 visits (-31.5%)** |
| **Gross A&E; Claims Cost** | S$ 2,940,000 | S$ 2,116,800 | **S$ 2,013,900** | **-S$ 926,100 saved** |
| **Diverted Telehealth Cost** | S$ 0 | S$ 68,600 | S$ 77,175 | +S$ 77,175 |
| **Net Medical Claims Savings** | — | S$ 754,600 | **S$ 848,925 / year** | **+12.5% uplift in deflection savings** |
| **Prevented ADE / Double-Dose Admissions** | 84 episodes (S$2,800 avg) | 18 episodes | **14 episodes** | **S$ 196,000 / year** |
| **Total Annualized Claims Reduction** | — | S$ 939,400 | **S$ 1,044,925 / year** | **+S$ 105,525 net incremental savings** |
```

2.3. Why TRTC Live Triage Exponentially Increases Insurer ROI

1. Higher Clinical Completion Rate (No Drop-Offs): In digital healthcare, form-based dropout rates exceed 45% when users encounter 4+ consecutive questions. With TRTC voice interaction, speech completion exceeds 88%, ensuring policyholders complete the full triage sequence rather than giving up and calling an ambulance.

2. Elderly & Multilingual Inclusion: Singapore policyholders over 65 represent the highest claim-frequency cohort. TRTC voice triage supports Singlish, Mandarin, and Malay, allowing seniors to voice their symptoms naturally without needing to type small fonts.

3. Instant Seamless Telehealth Escalation: When TRTC detects a Yellow/P2 or P3 urgency, the audio session can bridge directly into an insurer panel doctor tele-consult (e.g. WhiteCoat, Doctor Anywhere), preventing hospital visit leakage.

2.4. Commercial Pricing Models for Insurers

```
pie title Insurer Economic Value Distribution (Year 1 with TRTC)
    "Net Retained Insurer Claims Savings" : 52
    "GranPocket Enterprise Software & Voice Fee" : 35
    "Panel Telehealth Partner Integration" : 13
```

1. Model A: Fixed PEPM (Per Eligible / Covered Life Per Month)

- License Fee: S$1.50 PEPM (includes unlimited TRTC voice triage minutes, multi-agent reasoning, and medication safety locks).
- Annual Platform Investment: 50,000 × S$1.50 × 12 = S$900,000.
- Insurer Gross Claims Savings: S$1,044,925.
- Direct Insurer ROI: Break-even in Year 1 + S$144,925 immediate net surplus + 22% increase in customer Net Promoter Score (NPS).

2. Model B: Risk-Sharing Corridor (Recommended for Faster Anchor Signing)

- Base Platform Fee: S$0.75 PEPM (S$450,000/year).
- Shared Savings Success Fee: 25% of verified claims reduction below the agreed actuarial baseline.

- Total Verified Claims Reduction: S$1,044,925.
- Base Fee: S$450,000.
- Performance Fee: 25% × S$1,044,925 = S$261,231.
- Total GranPocket Enterprise Revenue: S$711,231.
- Net Insurer Hard Profit: S$333,694 net claims cash savings on day one.

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3. Anchor Partner Prioritization Matrix (Singapore Market)

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| Target Partner | Strategic Fit & Product Integration | Primary Decision Maker | Value Wedge with TRTC Live Triage |
| :--- | :--- | :--- | :--- |
| **1. AIA Singapore** *(Priority 1)* | **AIA HealthShield Gold Max** + **AIA Vitality** ecosystem. High mobile app engagement. | Head of Health & Claims / Chief Customer Officer | Embed as "AIA Live Voice Doctor" powered by TRTC. Anxious parents speak symptoms at 2 AM; deflects KKH A&E; claims; awards Vitality points for adherence. |
| **2. Great Eastern** *(Priority 2)* | **SupremeHealth** + **GREAT Family Care**. Strong brand equity in pediatric and critical illness. | Head of Group Healthcare Ecosystems | "GREAT GrandCare" Voice Assistant for elderly policyholders. Lowers geriatric polypharmacy admissions and infant fever claims. |
| **3. Prudential Singapore** *(Priority 3)* | **PRUShield** + **Pulse by Prudential** health app. Aggressive digital innovation focus. | Chief Digital Officer / Head of Medical Network | TRTC SDK embedded into Pulse app shell. Seamlessly hands off voice triage into Prudential's panel tele-doctor network. |
| **4. Singlife** *(Priority 4)* | **Singlife Shield**. Digital-first challenger insurer with agile tech integration. | Head of Digital Products & Partnerships | Rapid sandbox pilot (30 days). Digital-first branding showcasing Singapore's first conversational voice triage for IP policyholders. |
| **5. MOH / Healthier SG** *(Public Track)* | **HealthHub & National Electronic Health Record (NEHR)** integration. | Director, MOHT (MOH Office for Healthcare Transformation) | Universal voice interface for seniors who cannot read English or use complex app menus. Relieves polyclinic phone queues. |
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4. 180-Day Pilot Execution Blueprint

```
gantt
    title GranPocket 180-Day Anchor Insurer Pilot Roadmap (TRTC Enabled)
    dateFormat  YYYY-MM-DD
    section Phase 1: Setup & Cohort
    Sandbox Definition & Actuarial Baseline :2026-11-01, 30d
    section Phase 2: Technical Integration
    TRTC WebRTC Bridge & Panel Clinic Binding :2026-12-01, 30d
    section Phase 3: Clinical Pilot Run
    Dark Launch to 10k Family Pilot Cohort    :2026-12-31, 60d
    section Phase 4: Audit & Enterprise Rollout
    Claims Audit & Actuarial Attribution Review :2027-03-01, 30d
    Full 50k - 100k Commercial Enterprise Contract :2027-03-31, 30d
```

Stage-by-Stage Implementation

Phase 1 (Days 1–30): Cohort & Governance Setup

- Establish the Joint Steering Committee (Insurer Chief Medical Officer + GranPocket Clinical Governance Board).
- Identify pilot cohort: 10,000 policyholder families holding comprehensive hospital riders with at least 1 pediatric or geriatric dependent.
- Extract historical 24-month claims baseline for this specific cohort.
- Configure TRTC Cloud SDK credentials and Singapore edge clusters (`ap-singapore`) for compliance with MAS TRM and PDPA guidelines.

Phase 2 (Days 31–60): Technical & Claims Integration

- Implement OAuth2 / OIDC SingPass or Insurer App single-sign-on (SSO).

- Pre-populate panel clinics and telemedicine providers (e.g., WhiteCoat / Doctor Anywhere) within GranPocket's OneMap routing engine.
- Bind TRTC live voice audio pipeline to GranPocket's deterministic CPG rule engine and `triage_execution_logs` database audit trail.
- Conduct clinical red-teaming simulations verifying automatic emergency SOS interrupts on chest pain, stroke symptoms, and breathing distress.

Phase 3 (Days 61–120): Live Clinical Pilot Execution

- Launch GranPocket as an exclusive, complimentary benefit for the 10,000 policyholder families.
- Policyholders utilize TRTC conversational voice triage, photo wound assessments, and medication scheduling.
- Provide instantaneous virtual doctor escalation when Yellow-tier symptoms require prescription or physical clinic visit.

Phase 4 (Days 121–180): Actuarial Audit & Enterprise Contract Conversion

- Conduct formal claims variance audit comparing the pilot cohort against a matched control group.
- Measure verified A&E; deflection rate (targeting > 30%), net claims cost delta, and user satisfaction (CSAT > 92%).
- Convert pilot into a 3-year enterprise license contract across the insurer's entire policyholder base (250,000+ lives).

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5. Architectural Proof Points in Existing Codebase

GranPocket's existing codebase directly embodies the enterprise governance, real-time communication resilience, and clinical safety features demanded by institutional health payors:

1. Tencent RTC (TRTC) Live Voice Engine:

- Inspect [`server/trtcService.ts`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/server/trtcService.ts) and [`src/components/ConversationalTriageView.tsx`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/src/components/ConversationalTriageView.tsx): Ultra-low-latency real-time voice streaming with HMAC-SHA256 UserSig authentication, conversational state tracking, and red-flag audio interrupts.

2. Zero-Hallucination Tier 1 Clinical Rule Safety Net:

- Inspect [`src/engine/ruleBasedEngine.ts`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/src/engine/ruleBasedEngine.ts): Real-time voice inputs are grounded directly in CPG clinical decision trees (`getQuestionTree` and `runRuleBasedTriage`), guaranteeing models assist while rules decide.

3. Universal Triage Execution Audit Ledger (`triage_execution_logs`):

- Inspect [`server/db.ts`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/server/db.ts#L1530-L1565) and [`server/index.ts`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/server/index.ts#L1400-L1440): All triage sessions across Rule-Based, Agentic, and TRTC voice modes are permanently recorded with session IDs, answers, rule provenance, and deterministic trip statuses.

4. Physician Adversarial Attestation Loop:

- Inspect [`server/dynamicAgents.ts`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/server/dynamicAgents.ts#L1250-L1275): Every diagnostic draft is challenged by `@expert_doctor_agent` with up to 2 retry gates, preventing premature or dangerous self-care recommendations.

5. Cross-Caregiver Medication Safety Lock:

- Inspect [`src/components/MedicationLogView.tsx`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/src/components/MedicationLogView.tsx): Real-time synchronization across family members and domestic helpers with minimum dosing intervals preventing toxic double-dosing.

6. Institutional PDPA & Data Governance:

- Inspect [`src/components/CreateHouseholdModal.tsx`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/src/components/CreateHouseholdModal.tsx) and [`src/components/AuditAndPdpaView.tsx`](file:///Users/avijit/Documents/Avi/AI/Tencent%20Hackathon/CodeBuddy/GranPocket-Agenticv6_ADP%20TRTC/src/components/AuditAndPdpaView.tsx): Mandatory PDPA 2012 consent popup and immutable SHA-256 cryptographic audit ledger

tracking all clinical modifications.

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6. Investor Pitch Presentation Script (Slide-by-Slide with TRTC)

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SLIDE 1: THE S$100M EMERGENCY ROOM INFLATION TRAP
"In Singapore today, private health insurers are bleeding margin. Why? Because when a 3-year-old spikes a 38.5°C fever
SLIDE 2: MEET GRANPOCKET: THE MULTI-GENERATIONAL CLINICAL AIRBAG
"GranPocket is not a basic chatbot. It is an agentic clinical safety net built specifically for Singapore households.
SLIDE 3: PREVENTING THE UNSEEN KILLER: ACCIDENTAL DOUBLE-DOSING
"In homes where Mom, Dad, Grandma, and the domestic helper all care for the same child or elderly grandparent, who tra
SLIDE 4: THE ACTUARIAL ROI: HOW WE SAVE INSURERS OVER S$1,040,000 PER 50,000 LIVES
"We ran the numbers on Singapore Integrated Shield portfolios. By eliminating UX friction through Tencent RTC voice in
SLIDE 5: OUR GO-TO-MARKET ANCHOR: THE B2B2C STRATEGY
"We do not spend millions acquiring consumers on Google Ads. We partner directly with Singapore's tier-1 insurers—AIA,
SLIDE 6: THE ASK & 18-MONTH HORIZON
"We are raising S$1.2M in Seed funding to execute our 180-day anchor pilot with our first major Singapore insurer, sca
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